1 ـ الف

1. Start
2. Declaration Variables:
3. input Array [ ], n, k.
4. Start for_1 (i = 0 ; i < n ; update i = i+1) → then
5. Start for_2 (j = n+1 ; j < n ; update j = j+1)
6. multiply = Array[i] × Array[j]
7. if (multiply = k) ⇒ Print ("True")
8. End for_2.
9. Print ("False") (If program don't print ("True"))
10. End for_1.
11. End

۱ـ ب ـ

```
int checkMult( int Array [ ], int n, int k)
{
    for( int i = 0 ; i < n ; i++)
    { for(int j = i + 1 ; j < n ; j++) {
            int multiply = arr[i] * arr [j];
            if (multiply == k) {
                return 1;
            }
        }
    }
    return 0;
}
```

-2

```
int calculateBrightness(int image[10][10], int i, int j) {
    int Brightness Sum = 0;
    for (int z = i - 1; z <= i + 1; z++)
    {
        if (z < 0 || z >= 10)
        {
            continue;
        }

        for (int v = j - 1; v <= j + 1; v++)
        {
            if (v < 0 || v >= 10)
            {
                continue;
            }
            BrightnessSum += image[z][v];

        }
    }
    return BrightnessSum;
}
```

د -

```
void shiftRight (int arr[], int n, int k){
    int temp [n];
    for (int i=0; i<n; i++)
    {
        temp[i]= arr[i];
    }

 for ( int i=0 ; i<n ; i++)
  {
      if (i+k+1 >=n)
      {
        arr [i]= temp [i+k +1-n] ;
      }
      else
      { arr [i]= temp[i+1+k ];
      }
    }
}
```

روش ساده تری که بعداً به آن پی بردم:

```
void shiftRight (int arr[ ], int n, int k){
    int temp [n] ;
    for (int i=0 ; i<n ;i++)
     { temp[i]= arr [i];
     }

     for(int i=0; i<n ; i++){
       arr [(i+k)% n ] = temp [i ];
     }
 }
```

4

1- Start.

2- Declaration Variables: arr[], i=0, n, max, min.

3- Input n, arr[] n.

4- max = arr[0] & min = arr[0].

5- Start while ⇒ (i<=n-1)

6- if: (arr[i] > max) ⇒ max = arr[i].
   & if (arr[i] < min) ⇒ min = arr[i].

7- i = i+1.

8- Check if: i<=n-1 ⇒ Back to 5.

9- End while

10- Print ((max+min) div 2)

11- End

```
Void Dual_Inversion(char str[]){
    int i=0, charLength=0;
    while( Str[i] != '\0')
    {
        if ( Str[i] >= 'A' && Str[i]<= 'Z')
            str[i] += 32;

        else if ( Str [i] >= 'a' && Str[i] <= 'z')
            str [i] -= 32;

        CharLength +=1;
        i++;
    }

    for( int j=0 ; j< charLenght/2 ; j ++)
    {
        char temp= str [j];
        Str[j]= str [char Lenght - 1 -j]
        Str [CharLength- 1-j ] = temp;

    }
}
```